SMART PARKING PLATFORM

PROJECT MANAGEMENT PLAN

WE ARE $oftware ¢orp.

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1. Vision and Scope

WE ARE $oftware ¢orp. is developing the Smart Parking Platform as an internal commercial product. The web and mobile platform helps drivers find parking, reserve spaces, pay digitally and navigate to participating garages. Operators manage facilities, pricing, availability, reservations and reporting.

1.1 Scope

In scope	Out of scope
User accounts; live availability; parking search/map; facility details; reservations; digital payments; receipts/history; notifications; navigation integration; operator garage/pricing/inventory management; reporting.	Physical parking hardware; garage construction/operations; towing/enforcement; proprietary GPS infrastructure; equipment maintenance.

1.2 Stakeholders and Constraints

Stakeholders	Constraints
Drivers; parking operators; system administrators; city transportation departments; finance/billing staff; third-party payment providers.	Fixed semester delivery window; mapping/payment/data-service dependencies; privacy and security requirements; applicable city rules; unlimited theoretical development budget (with an explicit planning estimate).

2. Competitive Analysis

Platform	Existing capabilities	Project differentiation
SpotHero	Parking search, reservations, prepaid parking, operator tools.	Integrated availability-to-booking workflow and operator updates.
ParkWhiz	Parking reservations, payment and operator reporting/integrations.	Standardized facility data and workflow across garages.
ParkMobile	Reservations; parking management, pricing and analytics.	Shared real-time data for drivers and garage operators.

3. Software Requirements Specification

3.1 Functional Requirements

ID	Requirement
FR-01	Register/authenticate drivers and operators; logout and password recovery.
FR-02	Search parking by location, destination, date and time.
FR-03	Display parking facilities on an interactive map.
FR-04	Display facility hours, pricing, restrictions and available spaces.
FR-05	Show current availability from supported operator/data feeds; identify stale data.
FR-06	Create/cancel reservations and adjust inventory appropriately.
FR-07	Accept approved digital payments and handle failures safely.
FR-08	Provide booking confirmations, history and receipts.
FR-09	Notify users about reservation and account events.
FR-10	Open external supported mapping/navigation to a facility.
FR-11	Let authorized operators create, update or deactivate facilities.
FR-12	Allow operators to maintain parking inventory, rates and availability.
FR-13	Allow operators to review reservations and facility activity.
FR-14	Provide occupancy/utilization reporting across date ranges.
FR-15	Allow authorized finance users to view and export transaction reports.

3.2 Non-Functional Requirements

ID	Requirement
NFR-01	Encrypt data in transit and store passwords securely.
NFR-02	Use role-based access controls for driver, operator, admin and finance functions.
NFR-03	Target 99.9% service availability excluding maintenance and provider outages.
NFR-04	Target under 3 seconds for common requests excluding third-party delays.
NFR-05	Support responsive interfaces on mobile and web.
NFR-06	Support increasing user, facility and transaction volumes.
NFR-07	Log relevant administrative and financial actions for auditability.
NFR-08	Provide clear error handling and fallbacks during third-party outages.

3.3 User Stories

Story	As a... / I want to...	Acceptance check	SRS
US-01	Driver: create an account, sign in and recover access	Account validation, secure sessions and reset flow work.	FR-01
US-02	Operator: sign in to a role-appropriate workspace	Unauthorized users cannot access operator tools.	FR-01, 11
US-03	Driver: search parking near a destination for my trip time	Matching facilities respect entered location/date/time.	FR-02
US-04	Driver: see matching garages on a map and launch directions	Map markers open details; directions launch supported maps.	FR-03, 10
US-05	Driver: compare prices and facility restrictions	Selected facility presents rates, rules and availability.	FR-04
US-06	Driver: view current space availability	Latest supported feed is displayed; stale data is flagged.	FR-05
US-07	Driver: reserve an available parking space	Inventory is validated and a confirmation is created.	FR-06
US-08	Driver: cancel an eligible reservation	Status is updated and capacity is released appropriately.	FR-06
US-09	Driver: pay digitally and understand any failed payment	Approved payment recorded; failure avoids double booking.	FR-07
US-10	Driver: view past reservations and receipts	History links to the correct confirmation and receipt.	FR-08
US-11	Driver: receive reservation or account notifications	Relevant updates are delivered using chosen channel.	FR-09
US-12	Operator: create and edit parking facilities	Validated changes appear on authorized facility records.	FR-11
US-13	Operator: change parking rates and available capacity	Changes are validated and reflected in driver availability.	FR-12
US-14	Operator: review bookings and garage occupancy	Reservation activity and occupancy are reportable.	FR-13, 14
US-15	Finance staff: export reports of completed transactions	Only permitted users can filter/export transaction data.	FR-15

4. Use Cases

Use case	Actor	Main flow
UC-01 Register Account	Driver / Operator	Enter details; validate; create account.
UC-02 Log In	Driver / Operator / Admin	Enter credentials; validate; open authorized dashboard.
UC-03 Reset Password	Registered user	Request reset; verify identity; save new password.
UC-04 Search Parking	Driver	Enter location/date/time; return matching facilities.
UC-05 View Parking Map	Driver	Open map; display facilities; select a marker.
UC-06 Facility Details	Driver	Open facility; display price, rules and availability.
UC-07 Create Reservation	Driver	Select time; verify inventory; create booking.
UC-08 Pay for Reservation	Driver	Authorize payment; record transaction and receipt.
UC-09 Cancel Reservation	Driver	Select eligible booking; confirm; restore inventory.
UC-10 View History / Receipt	Driver	Select history item; show reservation and payment record.
UC-11 Navigate to Facility	Driver	Choose directions; hand off to navigation app.
UC-12 Add Facility	Operator	Enter facility details; validate and save.
UC-13 Edit Facility / Pricing	Operator	Edit details/rates; validate; save updates.
UC-14 Manage Availability	Operator	Update inventory; validate and refresh live feed.
UC-15 Review Reports	Operator / Finance	Select report/date range; aggregate/export data.

5. Work Breakdown Structure

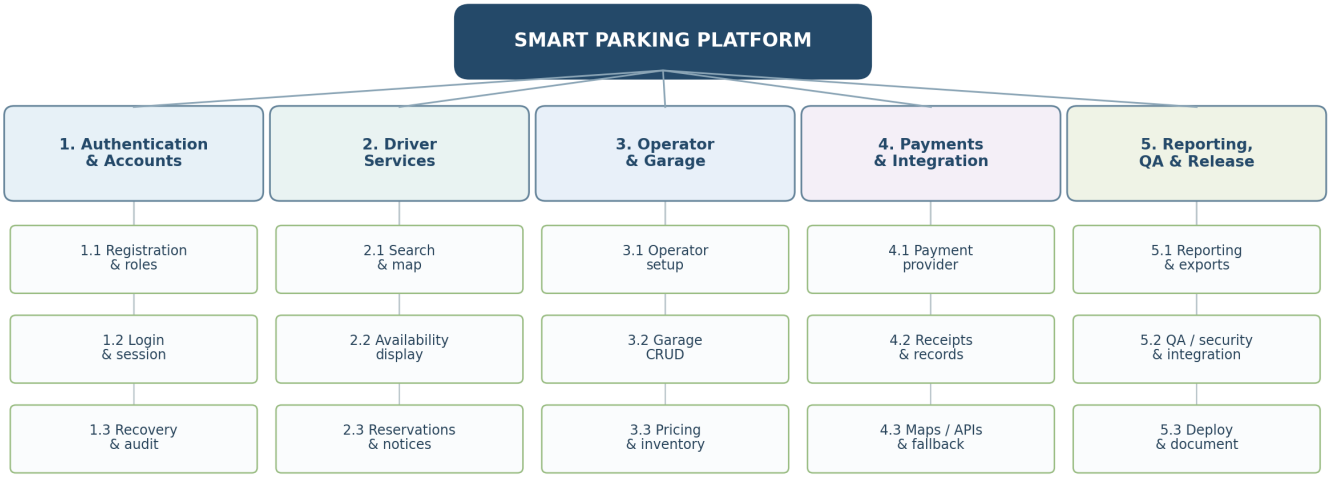


Figure 1. Work Breakdown Structure (three levels).

6. Project Schedule

Work package	Planned dates	Dependency	Planned output
Requirements baseline	Sep 01 - Sep 11	—	Requirements and scope reviewed
Architecture & UI design	Sep 08 - Sep 18	SS: requirements	System and screen design baseline
Authentication & accounts	Sep 14 - Sep 25	SS: architecture	Account and access workflows
Operator / garage management	Sep 21 - Oct 09	SS: authentication	Garage inventory/pricing workflows
Driver search, map & availability	Sep 28 - Oct 16	SS: operator services	Discovery and availability workflows
Reservations & payments	Oct 12 - Oct 30	SS: search; FS: core APIs	Booking and checkout workflows
Reporting & notifications	Oct 26 - Nov 06	SS: reservations	Operator/financial reporting
System integration	Nov 02 - Nov 13	SS: reporting; FS: core APIs	End-to-end integration evidence
QA & security testing	Nov 09 - Nov 27	SS: integration	Regression, security and fixes
Final release & documentation	Nov 23 - Dec 04	SS: QA; FS: release gate	Release package and documentation

FS = Finish-to-Start; SS = Start-to-Start.

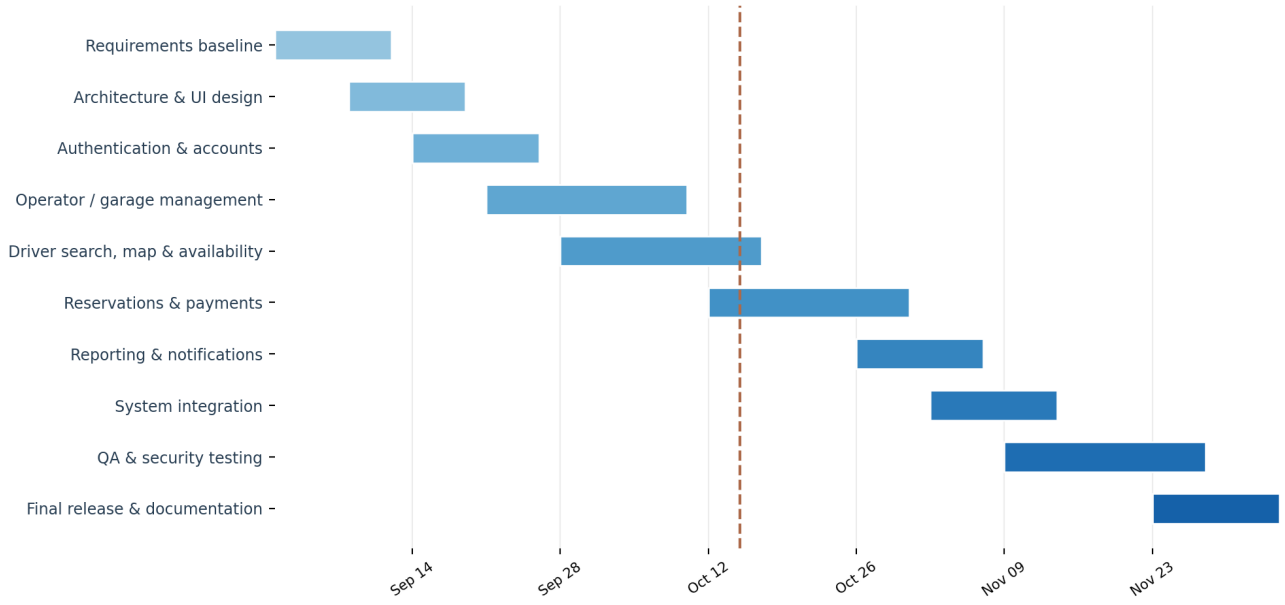


Figure 2. Planned semester schedule (dashed line: October 15 review date).

6.1 Milestones and Review Gates

Milestone	Planned date	Decision / evidence
Requirements baseline	Sep 11	Scope and requirement IDs confirmed.
Core operator work	Oct 09	Garage/inventory functions reviewed.
Driver search workflow	Oct 16	Map and availability review.
Reservations and payments	Oct 30	Booking and payment test walkthrough.
Integration readiness	Nov 13	End-to-end API and data checks.
QA/security acceptance	Nov 27	Critical defects and security findings assessed.
Final release	Dec 04	Approval, delivery package and documentation.

7. Product Backlog and Sprint 1

7.1 Login Backlog (5)

ID	Feature	Priority	Target
LOG-01	Driver Login	High	S3
LOG-02	Operator Login	High	S1
LOG-03	User Registration	High	S1
LOG-04	Password Reset	Medium	S2
LOG-05	Session Logout	Medium	S2

7.2 UI Backlog (15)

ID	Feature	Priority	Target
UI-01	Driver Dashboard	High	S3
UI-02	Operator Dashboard	High	S2
UI-03	Parking Search Interface	High	S2
UI-04	Interactive Parking Map	High	S2
UI-05	Parking Facility Details	High	S2
UI-06	Real-Time Availability Display	High	S2
UI-07	Reservation Screen	High	S3
UI-08	Payment Checkout Screen	High	S3
UI-09	Reservation Confirmation Screen	High	S3
UI-10	Reservation History Screen	Medium	S4
UI-11	Receipt View	Medium	S4
UI-12	Driver Profile Settings	Medium	S4
UI-13	Garage Management Interface	High	S2
UI-14	Pricing Management Interface	Medium	S2
UI-15	Operator Reporting Dashboard	Medium	S4

7.3 Backend Backlog (15)

ID	Feature	Priority	Target
BE-01	Driver Account Creation API	High	S2
BE-02	Operator Account Creation API	High	S1
BE-03	Authentication API	High	S1
BE-04	Password Recovery Process	Medium	S2
BE-05	Parking Facility Search API	High	S2
BE-06	Real-Time Availability Service	High	S1
BE-07	Garage Creation Process	High	S1
BE-08	Garage Update Process	High	S1
BE-09	Parking Inventory Management	High	S1
BE-10	Reservation Creation Process	High	S3
BE-11	Reservation Cancellation Process	High	S3
BE-12	Payment Processing Integration	High	S3
BE-13	Receipt Generation Process	Medium	S4
BE-14	Mapping & Navigation Integration	High	S3
BE-15	Notification Service	Medium	S4

7.4 Reporting Backlog (10)

ID	Feature	Priority	Target
REP-01	Current Occupancy Report	High	S4
REP-02	Daily Occupancy Report	Medium	S4
REP-03	Weekly Occupancy Report	Medium	S4
REP-04	Reservation Activity Report	Medium	S4
REP-05	Revenue Summary Report	High	S4
REP-06	Transaction History Report	Medium	S4
REP-07	Garage Utilization Report	Medium	S4
REP-08	Peak Parking Hours Report	Medium	S4
REP-09	Driver Reservation History Report	Medium	S4
REP-10	Financial Report Export	High	S4

7.5 Sprint 1 Backlog

ID	Sprint 1 item
LOG-02	Operator Login
LOG-03	User Registration
BE-02	Operator Account Creation API
BE-03	Authentication API
BE-06	Real-Time Availability Service
BE-07	Garage Creation Process
BE-08	Garage Update Process
BE-09	Parking Inventory Management

7.6 Planned Sprint Focus

Sprint	Planned window	Focus / deliverable
S1	Sep 14 - Sep 25	Authentication, operator account and garage/availability foundation.
S2	Sep 28 - Oct 09	Driver discovery, operator screens, recovery and garage workflows.
S3	Oct 12 - Oct 23	Reservations, payments, confirmations and navigation.
S4	Oct 26 - Nov 06	History, receipts, reports and notifications.
S5	Nov 09 - Nov 20	Integration fixes, functional, regression and security testing.
S6	Nov 23 - Dec 04	Release readiness, final QA gate and documentation.

7.7 Trello Board Evidence

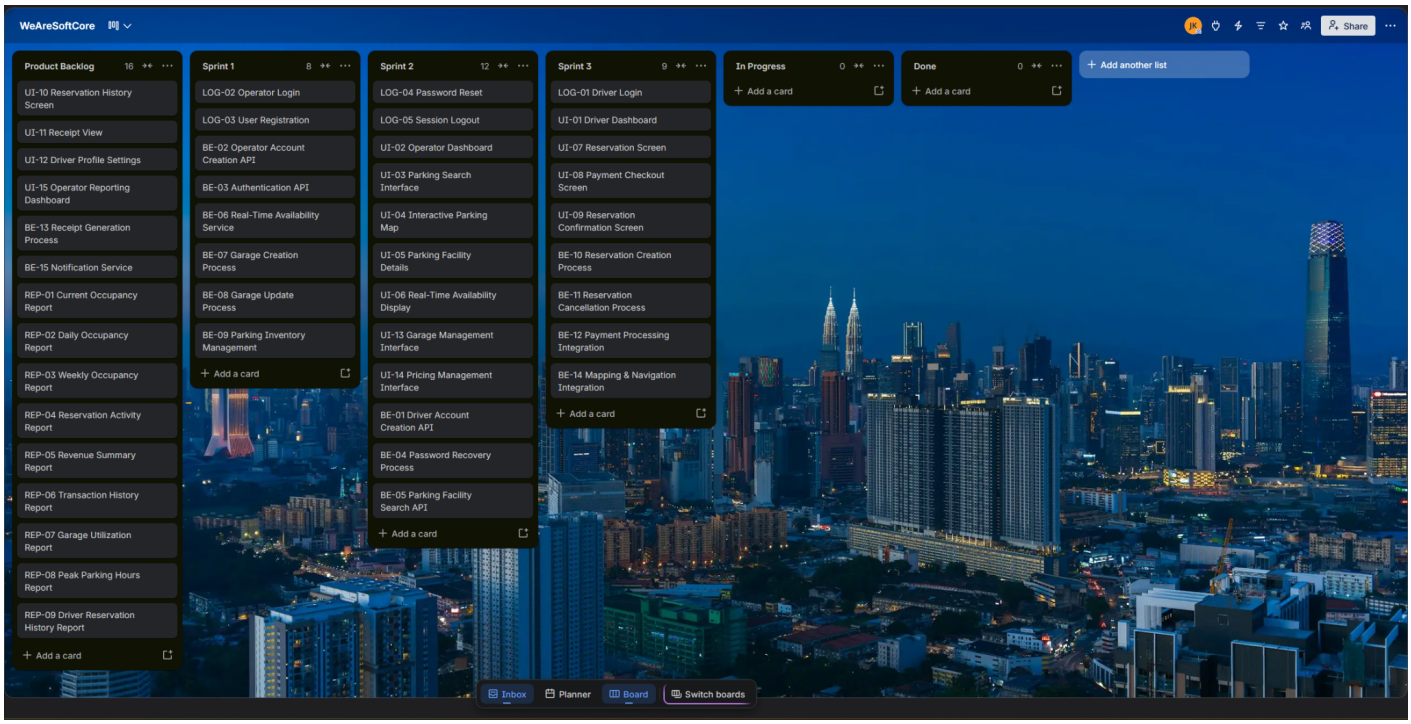


Figure 3. Updated Trello board: Product Backlog (16), Sprint 1 (8), Sprint 2 (12), Sprint 3 (9), In Progress (0), Done (0).

8. Risk Management

8.1 Technical Risks

ID	Risk description
T-01	Third-party mapping, payment or data APIs fail or change.
T-02	Parking availability feeds become outdated or inconsistent.
T-03	Authentication, payments or personal data face a security/privacy breach.
T-04	Frontend, backend and provider services fail to integrate reliably.

8.2 Schedule Risks

ID	Risk description
S-01	External-service integration takes longer than scheduled.
S-02	Authentication or garage work delays dependent driver features.
S-03	Late QA/security defects force major rework near the deadline.
S-04	Late scope changes displace final integration and documentation.

8.3 Financial Risks

ID	Risk description
F-01	Provider fees exceed budget estimates.
F-02	Rework increases planned labor cost.
F-03	Infrastructure costs grow faster than expected.
F-04	Compliance/security changes require unplanned spending.

8.4 People Risks

ID	Risk description
P-01	A key contributor becomes unavailable during a critical phase.
P-02	Skills gaps slow API, security or mobile/web implementation.
P-03	Miscommunication creates conflicting requirements and rework.
P-04	Uneven workload or ownership gaps delay delivery.

9. Risk Register

Probability and impact: 1-5 each. Risk score = probability × impact; Low 1-5, Medium 6-9, High 10-16, Extreme 17-25.

ID / Code	Risk	P	I	Priority	Owner	Response / control	Status
R-01 / T-01	Third-party API outage/change	3	4	High	Integration Lead	Mitigate; Monitor providers; retry, sandbox test and fallback.	Open
R-02 / T-02	Inaccurate real-time availability	3	4	High	Backend Lead	Mitigate; Validate time stamps; mark stale feeds; reconcile updates.	Open
R-03 / T-03	Security or privacy exposure	2	5	High	QA/Security Lead	Mitigate; Encrypt, enforce roles, review code and audit events.	Open
R-04 / S-01	External integration delay	3	4	High	Project Manager	Mitigate; Start sandbox work early; track blockers and buffers.	Open
R-05 / S-03	Late defects cause rework	3	4	High	QA/Security Lead	Mitigate; Start regression/security testing before final QA phase.	Open
R-06 / S-04	Scope growth or late changes	3	3	Medium	Product Owner	Avoid; Review changes; protect baseline and defer extras.	Open
R-07 / F-01	Third-party services cost more	2	3	Medium	Finance Lead	Mitigate; Review billing and usage; enable cost thresholds.	Open
R-08 / F-02	Rework increases labor expense	3	3	Medium	Project Manager	Mitigate; Agree acceptance checks; peer review; track defects.	Open
R-09 / P-01	Key contributor unavailable	2	4	Medium	Project Manager	Mitigate; Cross-train and document handoff; name backup.	Open
R-10 / P-03	Poor team communication	3	3	Medium	Project Manager	Mitigate; Capture decisions; weekly Trello review and escalation.	Open
R-11 / T-04	Cross-system integration defects	3	4	High	Integration Lead	Mitigate; Use contracts, staged integration and end-to-end checks.	Open
R-12 / S-02	Core work blocks driver services	3	4	High	Project Manager	Mitigate; Monitor dependencies; prioritize garage/auth blockers.	Open
R-13 / F-03	Unexpected cloud scaling costs	2	3	Medium	Finance Lead	Mitigate; Set utilization caps and check hosting usage.	Open
R-14 / F-04	Compliance/security cost change	2	4	Medium	QA/Security Lead	Mitigate; Review requirements early; use contingency only if approved.	Open
R-15 / P-02	Implementation skill gaps	3	3	Medium	Technical Lead	Mitigate; Pair specialists; time-box technical spikes.	Open
R-16 / P-04	Unbalanced work allocation	3	3	Medium	Project Manager	Mitigate; Review workloads and RACI owners each sprint.	Open

10. Communication Plan

Audience	Cadence	Participants	Method	Main content	Owner
Development Team	Daily	Developers; technical lead	Stand-up / chat	Blockers, priorities and dependencies	Technical Lead
Project / Sprint Team	Weekly	PM; PO; team leads	Meeting / Trello	Schedule, scope, sprint changes, open risks	Project Manager
QA & Security	Weekly; twice weekly during QA	QA/Security; developers	Defect/test review	Results, high-severity defects, release blockers	QA/Security Lead
Integration Team	Weekly + provider changes	Backend; integration	API checklist	Provider changes, contracts, fallback	Integration Lead
Stakeholders / Sponsors	Biweekly + gates	PM; PO; stakeholders	Report / demo	Milestones, risks and decisions required	Project Manager
Finance / Reporting	Monthly + pre-release	Finance; PM; PO	Cost review	Budget, API/cloud usage, reporting readiness	Finance Lead
Release Readiness	Before final gate	PM; PO; QA; technical	Go/no-go review	Defects, security, handoff and approvals	Project Manager

11. Roles and Resource Allocation

Role	Fictional name	Level	Allocation	Hours	$/hr	Cost
Project Manager	Alex Morgan	Senior	35%	168	$60	$10,080
Product Owner	Taylor Reed	Senior	20%	96	$55	$5,280
Technical / Backend Lead	Jordan Lee	Senior	75%	360	$65	$23,400
Frontend / UX Developer	Casey Nguyen	Mid-level	75%	360	$50	$18,000
Integration / DevOps Lead	Morgan Patel	Senior	50%	240	$65	$15,600
QA / Security Lead	Riley Chen	Mid-level	50%	240	$50	$12,000
Finance / Reporting Lead	Sam Rivera	Mid-level	20%	96	$45	$4,320

12. Resource and Cost Plan

12.1 Bottom-Up Cost Estimate

Category	Basis	Planned cost ($)
Labor	Staffing from Section 11	88,680
Cloud infrastructure	4 months × $350	1,400
Mapping/payment/notification APIs	4 months × $200	800
Licenses/developer tools	One-time allowance	600
Security/testing services	One-time allowance	800
Subtotal before contingency		92,280
Contingency reserve (10%)	10% of subtotal	9,228
Total planned budget		101,508

12.2 Budgeted Monthly Cashflow

Month	Labor ($)	Other ($)	Reserve ($)	Total ($)
September	17,736	1,150	1,846	20,732
October	31,038	550	3,230	34,818
November	26,604	1,350	2,768	30,722
December	13,302	550	1,384	15,236
TOTAL	88,680	3,600	9,228	101,508

13. RACI Matrix

Activity	PM	PO	TL	UX	INT	QA	FIN
Requirements / scope	A	R	C	C	I	I	I
Architecture / backend	C	I	A/R	C	C	C	I
Driver UI / search / map	I	C	C	A/R	C	C	I
Garage / availability services	I	C	A/R	C	R	C	I
Reservations / payments	I	C	A	R	R	C	C
System integration / deployment	C	I	R	C	A/R	C	I
QA and security testing	I	C	C	C	C	A/R	I
Reporting / financial export	I	C	R	C	C	C	A
Budget tracking / change control	A	C	C	I	C	I	R
Final release approval	R	A	C	I	C	C	I

R = Responsible; A = Accountable; C = Consulted; I = Informed. PM = Project Manager, PO = Product Owner, TL = Technical/Backend Lead, UX = Frontend/UX, INT = Integration/DevOps, QA = QA/Security, FIN = Finance/Reporting.